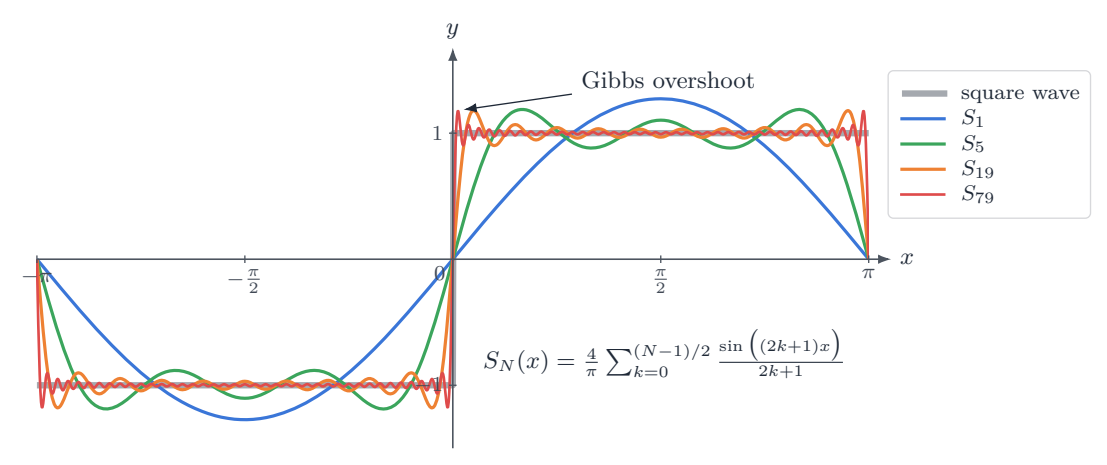

y
Gibbs overshoot
square wave
S_1
S_5
S_{19}
S_{79}
1
x
$-\pi$
$-\frac{\pi}{2}$
0
$\frac{\pi}{2}$
π
$S_N(x) = \frac{4}{\pi} \sum_{k=0}^{(N-1)/2} \frac{\sin\left((2k+1)x\right)}{2k+1}$